

Railway Data Cleaning Report

1. Overview

This report outlines the process of cleaning, transforming, and preparing the railway dataset for analysis. The structure has been revised to include a new fact table and supporting dimension tables that align with a well-defined star schema. A composite Journey_ID has been generated to uniquely identify journeys, enabling more robust analysis across time, location, and purchase details.

2. Star Schema Overview

Fact Table: FactSalesTickets

Contains transactional-level data including sales details and foreign keys linking to related dimension tables.

Key columns:

- Transaction ID
- Date and Time of Purchase
- Price
- Date of Journey
- Refund Request
- Ticket Class
- Foreign Keys: Purchase_ID, Discount_ID, Journey_ID

Dimension Tables:

DimJourney :- Stores journey-specific information to support analysis of travel patterns and delays.

Columns include:

- Journey_ID
- Departure Station
- Departure Time
- Arrival Destination
- Arrival Time
- Actual Arrival Time

- Reason for Delay
- Journey Status

DimCalendar :- Created from journey and purchase dates for time-based analysis.

Columns include:

- Date
- Day Name
- Day Number
- Month Name
- Month Number
- Year

DimPurchase :- Describes purchase characteristics.

Columns include:

- Purchase_ID
- Purchase Type
- Payment Method

DimDiscount :- Captures all discount-related details.

Columns include:

- Discount_ID
- Ticket Type
- Discount Ratio
- Railcard

3. Data Cleaning Steps

3.1 Missing and Redundant Data

- Removed columns not used in the star schema.
- Dropped duplicate rows from all tables.
- **Missing values in Actual Arrival Time and Reason for Delay were not filled or removed**, as their presence holds analytical meaning. These fields can help explain operational patterns such as on-time performance or reasons for journey disruptions.

3.2 Standardization and Consistency

- Unified inconsistent labels in categorical columns. Examples:
 - "Staff Shortage" → "Staffing"
 - "Signal failure" → "Signal Failure"
 - "Weather Conditions" → "Weather"
- Harmonized spelling and casing in station names and delay reasons.
- Applied consistent formatting to ticket class, railcard names, and purchase types.

3.3 Data Type Corrections

- Converted date fields to proper Date and Time format.
 - Ensured price and discount values were numerical.
 - Verified ID columns are in correct data types for relational joins.
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4. Journey_ID Generation Logic

To enable accurate grouping and joining of journey-related records, a custom Journey_ID was created using a **manual combination of fields** that define a unique trip.

Fields used to construct Journey_ID:

- Departure Station
- Arrival Destination
- Date of Journey
- Departure Time
- Arrival Time

These fields were concatenated manually into a single column to produce a consistent and human-readable journey identifier. This method ensured that each journey is uniquely represented, while avoiding unnecessary complexity in ID generation.

5. Outcome

The dataset is now fully cleaned and structured to support accurate analysis in Power BI and other tools. Key outcomes include:

- Cleaned and deduplicated fact and dimension tables.

- Standardized and validated data types.
- Integrated star schema with proper foreign key relationships.
- Retained meaningful missing values for further insight into delays.
- Created intuitive Journey_ID values to support reliable journey-level analysis.